Breast-CRAG: A Breast Cancer Large Language Model Leveraging Retrieval-Augmented Generation

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Abstract. The lack of specialized large language models (LLMs) for breast cancer limits their clinical adoption. To address this, we present Breast-CRAG, a 7B-parameter retrieval-augmented LLM specifically optimized for breast cancer applications. Our approach combines a fine-tuned generator (trained on a curated 268K-dialogue dataset) with a specialized retriever (leveraging a 1M-chunk knowledge base) to enhance response quality. Evaluations across four dialogue and two exam datasets demonstrate that Breast-CRAG outperforms comparable open-source models and achieves competitive performance with GPT-4o. Ablation studies confirm the contributions of both components. With its strong performance on clinical queries, Breast-CRAG represents a promising tool for breast cancer care and research.

Keywords: Large Language Model · Breast Cancer · Retrieval-augmented Generation

1 Introduction

The advancement of large language models (LLMs) presents opportunities for breast cancer care [1], yet clinical deployment faces challenges. While closed-source APIs [2] offer high performance with low computational demands, they raise privacy concerns due to cloud-based processing [3]. Conversely, open-source LLMs often exceed clinical computing capacities, and smaller models [4] (~10B parameters) suffer from inadequate humanization and specialization. These limitations highlight the need for lightweight, disease-specific LLMs in medical AI.

Current approaches limited by critical shortcomings: Prompt engineering adapts LLMs by carefully crafting instructions, but its performance can be inconsistent.[5] Retrieval-augmented generation (RAG) enhances LLMs with external knowledge, but building effective medical knowledge bases, particularly for specialized areas like breast cancer, remains a challenge.[6] Parameter-efficient fine-tuning (PEFT) optimizes a small

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subset of LLM parameters, but relies heavily on high-quality training data, which is scarce in many medical domains.[7].

Thus, we present Breast-CRAG, a retrieval-augmented LLM for breast cancer, combining a PEFT-tuned generator (trained on 268K dialogues) and a knowledge-aware retriever (1M chunks). It outperforms GPT-4o with only 7B parameters. Code/data can be accessed through https://github.com/Maxin-C/Breast-CRAG.

2 Method

As shown in Fig. 1, the Breast-CRAG generator constructed breast cancer datasets and trained an LLM. The MedDialog-CN [8] and Huatuo-26M [4] datasets were screened using keywords and GPT-4o was employed with a prompt to filter datasets. Then, we randomly sampled 2,000 instances for manual annotation by two human annotators and fine-tuned an LLM, which was then used to evaluate the remaining dataset, resulting in the Huatuo-26M-BC and MedDialog-BC. Furthermore, for the MedDialog-BC dataset, GPT-4o was used to condense multi-turn dialogues into single-turn Q&A pairs. By taking the doctor's response as a supervisory signal, we train the LLM using the instruction tuning.

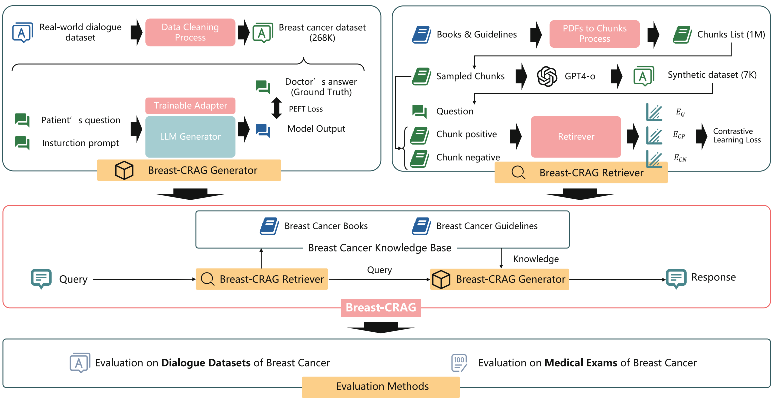

Fig. 1. Overview of Breast-CRAG Framework

In Breast-CRAG retriever training, the knowledge base was created from peer-reviewed books and guidelines (Chinese and English). The retriever was trained using a virtual dataset: GPT-4o generated Q&A pairs from knowledge chunks (positive samples), while random chunks served as negatives. A transformer encoder was trained via contrastive learning to align questions with positive samples.

Breast-CRAG was evaluated compared with GPT-4o [2] (released in 2024.08.06), HuatuoGPT2-7B [4] (HuatuoGPT2), and Qwen2.5-72B-Instruct-GPTQ-Int4 [9] (Qwen2.5-72B) on breast cancer dialogue dataset and exam dataset. The breast cancer dialogue dataset includes MedDialog-BC and Huatuo-BC test sets. To examine cross-dataset generalization ability, we included cMedQA [10] and webMedQA [11]. After process mentioned above, we got cMedQA-BC and webMedQA-BC. We evaluated models' output using Rouge, Bleu, and Bert-score. The breast cancer exam dataset comprises MedQA-USMLE [12] and web-scraped data from 2024 breast cancer medical licensing exams. After filtering for breast cancer and removing language overlaps, we obtained the USMLE-BC dataset and created ExamBC dataset from web-scraped data. Based on whether the answer requires integration of patient medical history information, we divided the dataset into a simple set and a hard set. Models were evaluated on dataset with exam prompt. Single-choice accuracy is the correct-answer proportion; multiple-choice scores 1 (fully correct), 0.5 (partially correct without errors), or 0, with overall accuracy averaged across the dataset.

3 Result

3.1 Main Result

Breast-CRAG generator was fine-tuned on 91K dialogue dataset based on Qwen2.5-7B-Instruct [9] (Qwen2.5-7B) and Breast-CRAG retriever was trained on 7K synthetic dataset based on nomic-embed-text-v1.5 [13].

As shown in Table 1, with temperature set to 0.01, Breast-CRAG achieved the best evaluation results on Huatuo-BC and MedDialog-BC, as well as the unseen datasets cMedQA-BC and webMedQA-BC.

Table 2 presents the results. GPT-4o generally outperformed other models, while Breast-CRAG achieved comparable performance, with minor gaps.

3.2 Ablation Study

Ablation studies' results (Table 3) across all three datasets revealed similar trends: both the Breast-CRAG generator and retriever positively impacted model performance. On the Huatuo-BC test set, the generator had a more pronounced effect. On the Exam-BC dataset, both components contributed equally.

4 Discussion

Breast-CRAG overperformed GPT-4o's performance on breast cancer dialogues dataset, and surpassed other open-source LLMs. This outcome supports our argument that smaller LLMs trained on medical domain-specific data and knowledge can outperform larger models—especially when resources are focused on specialized fields, enabling optimal results in specific applications at a lower cost.

Breast-CRAG showed improvements on USMLE and Exam-BC, nearing GPT-4o's performance. However, like other LLMs, it struggled with multiple-choice questions, often selecting "all of the above," possibly due to retrieval interference errors.

Both the generator and retriever contributed, with the generator having a larger impact by embedding breast cancer knowledge in parameters. The retriever, while not boosting interpretability significantly, constrained generation scope, aiding clinical applicability.

Table 1. Results on breast cancer dialogue dataset, where Bleu are presented as percentages

Detaset	Model	Rouge-1	Rouge-2	Rouge-L	Bleu	Bert-score
Huatuo-BC (test set)	HuatuoGPT2	0.167	0.025	0.110	2.492	0.835
	Qwen2.5-72B	0.158	0.023	0.096	2.255	0.825
	GPT-4o	0.155	0.021	0.084	1.865	0.819
	Breast-CRAG	**0.228**	**0.057**	**0.191**	**5.052**	**0.873**
MedDialog-BC (test set)	HuatuoGPT2	0.227	0.050	0.124	3.767	0.828
	Qwen2.5-72B	0.203	0.044	0.095	2.992	0.814
	GPT-4o	0.189	0.037	0.080	2.314	0.813
	Breast-CRAG	**0.230**	**0.052**	**0.191**	**5.277**	**0.876**
cMedQA-BC	HuatuoGPT2	0.158	0.023	0.102	2.108	0.829
	Qwen2.5-72B	0.141	0.018	0.092	1.770	0.823
	GPT-4o	0.151	0.020	0.082	1.644	0.816
	Breast-CRAG	**0.233**	**0.059**	**0.198**	**5.014**	**0.874**
webMedQA-BC	HuatuoGPT2	0.161	0.023	0.106	2.776	0.819
	Qwen2.5-72B	0.153	0.019	0.094	2.276	0.813
	GPT-4o	0.155	0.021	0.085	2.038	0.808
	Breast-CRAG	**0.182**	**0.043**	**0.153**	**3.272**	**0.844**

Table 2. Results on breast cancer exam dataset, where Multi. Refers to multiple-choice questions and Ave. Represents the average score across the dataset

Models	Exam-BC (simple set)			Exam-BC (hard set)			USMLE-BC
	Single	Multi.	Ave.	Single	Multi.	Ave.	Single
HuatuoGPT2	0.36	0.36	0.36	0.26	0.24	0.25	0.48
Qwen2.5-7B	0.35	0.44	0.37	0.30	**0.49**	0.33	0.41
GPT-4o	**0.73**	**0.54**	**0.69**	**0.68**	0.45	**0.64**	**0.81**
Breast-CRAG	0.65	**0.54**	0.63	0.56	0.37	0.52	**0.81**

The limitations of this study are twofold. Breast-CRAG is text-only, missing multi-modal data in PDF conversion. It also lacks real-world clinical testing and physician evaluations. Expanding modalities and clinical integration, we plan to deploy Breast-CRAG in outpatient settings for endocrine therapy management, using EHR data ethically to enhance patient care and adherence.

Table 3. Ablation studies' results on Huatuo-BC (test set), Exam-BC and USMLE-BC, where w/o Breast-CRAG generator refers to using Qwen2.5-7B-Instruct without PEFT and w/o Breast-CRAG retriever refers to using the untuned nomic-embed-text-v1.5 as the retriever

Model	Rouge-L	Bleu	Bert-score	Exam-BC (simple)	Exam-BC (hard)	USMLE-BC
Base model	0.096	2.265	0.824	0.37	0.33	0.41
w/o generator	0.125	2.972	0.833	0.60	0.48	0.53
w/o retirever	0.162	4.221	0.857	0.61	0.48	0.78
Breast-CRAG	**0.198**	**5.052**	**0.873**	**0.63**	**0.52**	**0.81**

5 Conclusion

We developed Breast-CRAG, a breast cancer-specific LLM with an automated data/knowledge pipeline, featuring a specialized generator and retriever. It outperformed on benchmarks, offering a deployable solution for low-resource clinical settings and insights for other medical LLMs. Future work will improve modality compatibility and clinical applicability to tackle real-world challenges.

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